

Query Answer

AI Language Model Prompter

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To solve the quadratic equation $x^2 - 2x - 3 = 0$ and express the solution in a valid JSON string format, we can follow these steps:

1. Use the quadratic formula^{**}: The quadratic formula is given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Plugging in the values of a , b , and c , we get:

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot (-3)}}{2 \cdot 1} = \frac{2 \pm \sqrt{4 + 12}}{2} = \frac{2 \pm \sqrt{16}}{2} = \frac{2 \pm 4}{2}$$

This gives us two solutions:

$$x = \frac{2 + 4}{2} = \frac{6}{2} = 3$$

and

$$x = \frac{2 - 4}{2} = \frac{-2}{2} = -1$$